Columbia Econ Math Camp 2025

Real Analysis

Bruno Escobar

August 11, 2025

1. Sets, Relations and Functions

Sets*

A **set** is a grouping of objects characterized by its **elements** or **members**.

- We denote x *belongs to/is an element of* the set X as $x \in X$.
- Sometimes we will define sets through exhaustive listing of its elements, always using curly brackets:
  $X = \{1, \pi, \text{apple}, \text{Economics}, \text{Columbia University}\}$.
- Other times we will write a suggestive listing of its elements, such as $X = \{1, 2, 3, \cdots\}$.
  The set here has infinitely many elements so an exhaustive list is unfeasible; however, it is "obvious" that we are adding 1 to each subsequent element.
- More commonly, we write a set starting by their typical elements and their notation followed by properties that its elements must satisfy:
  $X = \{\text{integers s.t. their square is } 4\} = \{n \in \mathbb{Z} : n^2 = 4\} = \{-2, 2\}$.

Sets*

Definition 1 (Inclusion)

We say that A is **included in** B, or that A is a **subset** of B, and denote $A \subseteq B$ (or $A \subset B$) if every member of A is a member of B: $x \in A \Rightarrow x \in B$.

- Any set is a subset of itself.
- VERY IMPORTANT: We denote $A = B$ iff $A \subseteq B$ and $B \subseteq A$.
- We denote $A \underset{\neq}{\subseteq} B$ but $A \neq B$ iff $A \subset B$ but $B \subset A$ is not true. In this case, we say that A is a *proper subset* of B.
    - A lot of textbooks use $\subset$ to denote proper subsets; bear that in mind.
- The inclusion is *transitive*: if $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.
- The set of all subsets of a set X is called the *power set* of X, and noted $P(X)$ or 2^X.
    - The latter gets its name because if X has n elements, then $P(X)$ has 2^n elements.

Sets*

Definition 2 (Intersection and Union)

Consider two sets A and B. Their **intersection** is the set of all elements that *are both* in A and B:

$$A \cap B := \{x : x \in A \text{ and } x \in B\}$$

Their **union** is all the elements that *are either* in A or B (or both).

$$A \cup B := \{x : x \in A \text{ or } x \in B\}$$

- Both are commutative: $A \cup B = B \cup A$, and $A \cap B = B \cap A$.
- Both are associative: $(A \cap B) \cap C = A \cap (B \cap C)$, and $(A \cup B) \cup C = A \cup (B \cup C)$.
- $A \cap B = A$ iff $A \subset B$, and $A \cup B = A$ iff $B \subset A$.

Sets*

Generalize the notation we defined previously to allow for infinite (and possibly uncountable) union/intersection:

$$\bigcup_{\theta \in \Theta} A_\theta := \{x : \exists\, \theta \in \Theta \text{ s.t. } x \in A_\theta\}$$

$$\bigcap_{\theta \in \Theta} A_\theta := \{x : x \in A_\theta, \forall \theta \in \Theta\}$$

- The intersection is distributive w.r.t. the union:

$$B \cap \left(\bigcup_{\theta \in \Theta} A_\theta \right) = \bigcup_{\theta \in \Theta} (B \cap A_\theta).$$

- The union is distributive w.r.t. the intersection:

$$B \cup \left(\bigcap_{\theta \in \Theta} A_\theta \right) = \bigcap_{\theta \in \Theta} (B \cup A_\theta).$$

Sets*

- Sets (together with relations and functions) are the foundational objects of logic and mathematical anaylsis.
  - To put simply, *everything* in maths can be thought of as a set.
- When doing proofs, it will be very useful if you are comfortable with thinking in terms of (and handling) sets.
- A very important set is the one/unique set that contains no elements. We call this the **empty set**, and denote it by $\emptyset$.
  - An **axiom** in logic is that $\emptyset \subset A$ for any set A that you can ever make.
  - An equivalent axiom are that, for any set A, $\emptyset$ is the only set such that $A \cap \emptyset = \emptyset$ and $A \cup \emptyset = A$.
  - In proof-making, this set is useful to convey the idea that "no object satisfies so-and-so property".

Sets*

Definition 3 (Difference)

Given two sets A and B, the **set difference** $A - B$ or $A \setminus B$ is the set of all x that belong to A but not to B:

$$A - B := A \setminus B := \{x \in A : x \notin B\}$$

- Suppose that elements of A all naturally belong to a higher supra-set U (e.g. A is the set of odd integers, so U are all integers) so $A \subset U$. Then the **complement** of A is defined as

$$A^c := U \setminus A$$

- We note $x \notin A$ for $x \in A^c$ (i.e., $x \in U$ implicitly).
- $(A^c)^c = A$.

Sets*

Theorem 1 (De Morgan's Law)

Let $\{A_\theta\}_{\theta\in\Theta}$ *be a collection of subsets of* U*. We have:*

$$(1)\quad \left(\bigcup_{\theta\in\Theta} A_\theta\right)^c = \bigcap_{\theta\in\Theta} A_\theta^c$$

$$(2)\quad \left(\bigcap_{\theta\in\Theta} A_\theta\right)^c = \bigcup_{\theta\in\Theta} A_\theta^c$$

Exercise 1

Prove De Morgan's Law.

Sets*

Definition 4 (Cartesian Product)

The **Cartesian product** of A and B is the set of all ordered pairs (a, b) whose first element a is taken from A and whose second element b is taken from B. We denote it as

$$A \times B := \{(a, b) : a \in A \text{ and } b \in B\}$$

- The pair is ordered in the sense that (a, b) is different from (b, a).
- We can also talk about the Cartesian product of multiple sets $A_1 \times A_2 \times \cdots \times A_n$, or simply

$$\prod_{i=1}^{n} A_i := \{(a_1, a_2, \ldots, a_n) : a_i \in A_i, \forall\, n = 1, 2, \ldots, n\}$$

- If all A_i's are the same, order does not matter and we can denote $A \times \cdots \times A$ as A^n (e.g. $\mathbb{R}^n$).

Relations

- A **(binary) relation** R **from** A **to** B is simply a subset of $A \times B$.
  - Any comparison operation or symbol you can think of is formally a relation: $>, \geq, <, \leq, =, \neq$,etc.
  - For example $\leq := \{(x, y) \in \mathbb{R}^2 : y - x \text{ is non-negative}\}$ (on $\mathbb{R}$).
  - It is obvious $1 \leq 2$ is also $(1, 2) \in \leq$.
  - Whereas clearly it is not true that $2 \leq 1$ or $(2, 1) \notin \leq$.
- For any $(a, b) \in A \times B$ either if $(a, b) \in R$ or $(a, b) \notin R$.
  - This is not the same as completeness!

  For a relation R from A to B, its inverse is a relation from B to A, defined as
  $$R^{-1} := \{(b, a) \in B \times A : (a, b) \in R\}$$
  - We can always invert a relation unlike functions (e.g. $\geq = \leq^{-1}$).
- A relation R from A to A itself is also called a **relation on** A.
  - The inclusion relation $\subset$ of sets can be viewed as a relation on the set of all sets.

Relations

Definition 5 (Relations axioms)

Let R be a relation on the set X. We say R is ...

1. ... **reflexive** iff xRx for any $x \in X$.
2. ... **transitive** iff for any $x, y, z \in X$ such that xRy and yRz, we have xRz.
3. ... **anti-symmetric** iff for any $x, y \in X$ such that xRy and yRx, we have $x = y$.
4. ... **complete** iff for any $x, y \in X$, either xRy or yRx.
5. ... **symmetric** iff for any $x, y \in X$ such that xRy, we have yRx.

Exercise 2

Show that completeness implies reflexiveness.

Relations

Definition 6 (Pre-order)

A relation $\leq$ on X is a **pre-order** iff $\leq$ is reflexive and transitive. A pre-order on X is further called an **equivalence relation** if in addition it is symmetric; we usually denote it as $\sim$.

- If $\leq$ is a pre-order, we often define $a < b$ as ($a \leq b$ and $a \neq b$).
- The usual *less than* relation is a pre-order on $\mathbb{R}^n$.
- The classic economic preference $\succsim$ (as you will see in micro) is fundamentally a pre-order on a set of goods/bundles.

Relations

Definition 7 (Ordered Sets)

A relation $\leq$ on X is a **partial order** iff $\leq$ is reflexive, transitive, and anti-symmetric. In this case, we call $(X, \leq)$ a **partially ordered set**, or a **poset**. If the relation is in addition complete, then $(X, \leq)$ is a *totally ordered set*.

- $(\mathbb{R}, \leq)$ is a totally ordered set.
- $(\mathbb{R}^n, \leq)$ is a poset, but in general it is not totally ordered for $n > 1$.
- The classic economic preference $\succsim$ is not a poset on most economic sets of bundles since a partial order precludes *ties* between two objects.

Bounds

Definition 8 (Bounds)

Let $(X, \leq)$ be a poset, and let $A \subset X$.

- $u \in X$ is an **upper bound** of A iff $u \geq x$, $\forall x \in A$. If such u exists, we say that A is bounded from above.
- $l \in X$ is a **lower bound** of A iff $l \leq x$, $\forall x \in A$. If such l exists, we say that A is bounded from below.

- A set is said to be **bounded** if it has both upper and lower bound(s).
- If A is the empty set, then any $u \in X$ is an upper bound, since the requirement ($u \geq x$ for any $x \in A$) is void.
- A set may have multiple or even infinitely many upper/lower bounds. However we are most interested in one specific upper (lower) bound: their supremum (infimum).

Bounds

Definition 9 (Supremum/Infimum)

Let $(X, \leq)$ be a poset, and let $A \subset X$.

1. $u \in X$ is the *least upper bound*, or *supremum*, of A iff
   (a) u is an upper bound of A, and
   (b) $u \leq v$ for any upper bound v of A.
2. $l \in X$ is the *greatest lower bound*, or *infimum*, of A iff
   (a) l is a lower bound of A, and
   (b) $l \geq m$ for any lower bound m of A.

We denote this as $u = \sup A$ and $l = \inf A$

Exercise 3

In the standard $(\mathbb{R}, \leq)$ poset, show that 0 is the infimum of $A = \{1/n : n \text{ is natural}\}$.

Bounds

- Even if there are upper bounds, supremums may not exist for a set, e.g. $X = \mathbb{R} \setminus \{0\}$ and $A = \{x \in \mathbb{R} : x < 0\}$.
- But if they exist, supremum/maximum must be unique.
- A poset $(X, \leq)$ where any bounded subset from above has a least upper bound is said to have the **l.u.b. property**.
- The defining property of the real line $\mathbb{R}$ is that satisfies l.u.b. property.
- This is unlike, say, the rational numbers $\mathbb{Q}$, which do not satisfy l.u.b. property, e.g. $A = \mathbb{Q} \cap [0, \sqrt{2}]$ does not have l.u.b. in $\mathbb{Q}$.

Exercise 4

Show that the supremum of a set (should it exist) is unique (Hint: Use anti-symmetry).

Bounds

Definition 10 (Maximum/Minimum)

Let $(X, \leq)$ be a poset, and let $A \subset X$.

1. $x^* \in A$ is a maximum of A iff x^* is a upper bound of A.
2. $x_* \in A$ is a minimum of A iff x_* is a lower bound of A.

- As with supremum, should it exist, maximums are unique.
- The crucial difference between the maximum and supremum of A is that maximums must be elements of A itself.

Proposition 1

Let $(X, \leq)$ be a poset, and let $A \subset X$.

1. If a maximum of A exists, then it is the supremum of A.
2. If a minimum of A exists, then it is the infimum of A.

Functions

Definition 11 (Function)

A relation f from X to Y is a **function** or **mapping** iff

(1) $\forall x \in X, \exists y \in Y$ such that $(x, y) \in f$, and

(2) $\forall x \in X$ and $y_1, y_2 \in Y$ such that $(x, y_1) \in f$ and $(x, y_2) \in f$, we have $y_1 = y_2$.

- (1) and (2) imply that for each x in X, there exists a unique y in Y such that $(x, y) \in f$.
- Thus we can unambiguously write $y = f(x)$, or that y is the value of f evaluated at x, instead of using the relation notation $(x, y) \in f$.
- Similarly we denote a function f from **domain** X to **codomain** Y as $f : X \to Y$.

Functions

Definition 12 (Image)

The **image** of a set $S \subset X$ under f is the set of all $y \in Y$ for which some $x \in S$ satisfies $f(x) = y$, i.e.

$$Im(S) := f(S) := \{y \in Y : y = f(x) \text{ for some } x \in S\} = \bigcup_{x \in S} \{f(x)\}$$

- The **inverse image** (or **pre-image)** of a set $T \subset Y$ under f is

$$f^{-1}(T) := \{x \in X : f(x) \in T\}$$

- The image of the whole domain X under f, denoted $f(X)$ or $Im(X)$, is called the **range** of f.
- Notice that $f(X)$ may be a proper subset of the codomain Y, since there may exist y's that correspond to no x.

Functions

Proposition 2

Let $f : X \to Y$, $S_1, S_2 \subset X$ and $T_1, T_2 \subset Y$.

(1a) $S_1 \subset S_2 \;\Rightarrow\; f(S_1) \subset f(S_2)$

(1b) $T_1 \subset T_2 \;\Rightarrow\; f^{-1}(T_1) \subset f^{-1}(T_2)$

(2a) $f(S_1 \cup S_2) = f(S_1) \cup f(S_2)$

(2b) $f^{-1}(T_1 \cup T_2) = f^{-1}(T_1) \cup f^{-1}(T_2)$

(3a) $f(S_1 \cap S_2) \subset f(S_1) \cap f(S_2)$

(3b) $f^{-1}(T_1 \cap T_2) = f^{-1}(T_1) \cap f^{-1}(T_2)$

(4) $f^{-1}(T^c) = \left(f^{-1}(T)\right)^c$

Exercise 5

Prove (1a), (1b), (2a), (3b), (4) at home. I will do the rest in class.

Functions

Definition 13 (Composition)

Let $f : X \to Y$ and $g : Y \to Z$. Then the **composition** of f and g, denoted as $g \circ f$, is a function from X to Z such that for each $x \in X$, the value of the function $g \circ f$ is defined as

$$(g \circ f)(x) := g(f(x)).$$

- Composition is associative: $(h \circ g) \circ f = h \circ (g \circ f)$.
- However it is not commutative, i.e. we cannot switch the order of h, g, and f.

Functions

Definition 14 (Injection, surjection, and bijection)

Consider a function $f : X \to Y$.

1. f is an **injective function**, or *injection*, iff $\forall\, x_1, x_2 \in X$ such that $f(x_1) = f(x_2)$, we have $x_1 = x_2$.
2. f is a **surjective function**, or *surjection*, iff $f(X) = Y$.
3. f is a **bijective function**, **bijection**, or **invertible function** iff f is both injective and surjective.

- Injection requires that each y corresponds to at most one x.
- Surjection requires that each y corresponds to at least one x.
- Bijection requires that each y corresponds to exactly one x.

Exercise 6

Show that inverse relation of f, f^{-1}, is a function iff f is a bijection.

Functions

When both the domain and the codomain are ordered sets, we can talk about monotonicity of a function.

Definition 15 (Montonocity)

Let $(X, \leq_X)$ and $(Y, \leq_Y)$ be posets, and consider a function from X to Y.

1. f is **weakly increasing** iff $x \leq_X x' \Rightarrow f(x) \leq_Y f(x')$.
2. f is **weakly decreasing** iff $x \leq_X x' \Rightarrow f(x) \geq_Y f(x')$.
3. f is **strictly increasing** iff $x <_X x' \Rightarrow f(x) <_Y f(x')$.
4. f is **strictly decreasing** iff $x <_X x' \Rightarrow f(x) >_Y f(x')$.

Intuitively, a monotonic function is one that preserves, or inverses, the order in the domain.

2. Numbers*, Countability and Cardinality**

Numbers*

- The **naturals** are $\mathbb{N} = \{1, 2, 3, 4, \dots\}$.
- The **integers** add the zero and the negative of all naturals $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$.
- The **rationals** are the set of all possible divisions between two integers, so $\mathbb{Q} = \{m/n : m, n \in \mathbb{Z}, n \neq 0, m \text{ and } n \text{ are co-prime}\}$. (You can omit the co-prime property, it just adds redundancy.)
- The **real** numbers $\mathbb{R}$ are the union of the rationals and **irrationals** - the latter are those that cannot be written as division of two integers.
  - If you want, you could wlog represent real numbers through sequences:
    - $2 \to s = (\dots, 0, 0, 2, 0, 0, 0, \dots)$.
    - $10.5 \to s = (\dots, 0, 1, 0, 5, 0, 0, \dots)$.
    - $\pi \to s = (\dots, 0, 0, 3, 1, 4, 1, \dots)$.
  - Crucially, the real number line satisfies the l.u.b. property.
  - Moreover, any open interval in $\mathbb{R}$ contains infinitely many rational and irrational numbers; we call this that $\mathbb{Q}$ are **dense** in $\mathbb{R}$.
  - Also R_+ denotes non-negatives, R_{++} denotes positives.

Numbers*

- The **complex** numbers, which can always be represented as $z = a + bi$ where $i = \sqrt{-1}$, are at their core the same as $\mathbb{R}^2$. We call a the **real part** of z, or $Re(z)$, and b the **imaginary part** or $Im(z)$.
- They arise when solving the roots of real polynomials, so handling them is useful particularly in time-series or macro-models.
- e.g. In $P(x) = x^2 + 4$, the polynomial P has roots $\pm 2i$.
- The **conjugate** of $z = a + bi$ is $\bar{z} = a - bi$. A famous theorem states that if z is root of polynomial P, then $\bar{z}$ must also be a root.
- The **modulus** is a measure of how big the complex number is in the (Re, Im) plane, and its formula is simply the Pythagoras Theorem: $|z| = \sqrt{a^2 + b^2}$.
  - It generalizes the notion of **absolute value** in the real line, as a real number is simply a complex with zero imaginary part!

Countability**

Definition 16 (Countability)

A set X is **countably infinite** if there exists a bijection between $\mathbb{N}$ and X. We say X is **countable**, then, if it is finite or countably infinite, and **uncountable** otherwise. Equivalently, X is countable if there exists an injection of X onto $\mathbb{N}$.

- Intuitively, an infinitely large set is countable if you can index it discretely: $X = \{x_1, x_2, x_3, \ldots\}$
- Obviously, under this definition $\mathbb{N}$ is trivially countable using the identity function $f(i) = i$, where $i \in \mathbb{N}$.
- But we can use that to show, say, that also $\mathbb{Z}$ is countable.

Exercise 7

Show that $\mathbb{Z}$ is countable.

Countability**

Proposition 3

If X is countably infinite then any infinite subset $Y \subset X$ is also countably infinite.

To see this, wlog note that $Y = \{y_1, y_2, y_3, \dots\} = \{x_{n_1}, x_{n_2}, x_{n_3}, \dots\}$ for some bijection $i \mapsto n_i$.

Proposition 4

A countable union of countable sets is countable.

For a countable collection of sets $\{S_1, S_2, S_3, \dots\}$, where $S_j = \{x_1^j, x_2^j, x_3^j, \dots\}$ we can map $f(x_i^j) = 2^i 3^j$. This function is clearly injective (each $f(x_i^j)$ is a different number) so it must be countable. You can also order the numbers $\{2^i 3^j\}_{i,j \in \mathbb{N}}$ from smallest to largest to find a bijection between $(S_1, S_2, S_3, \dots)$ and $\mathbb{N}$.

Countability**

Because of the previous two proposition, a lot of famous results on countability follow immediately.

Proposition 5

$\mathbb{Q}$ is countable.

Proposition 6

The Cartesian product of finitely many countable sets is countable.

Exercise 8

Prove that $\mathbb{Q}$ is countable (Hint: Look at the last proof from the previous slide).

Cardinality**

Definition 17 (Cardinality)

Let S and T be two sets.

- S and T have *equal cardinality* if there exists a bijection between them, or $|S| = |T|$
- S has *higher cardinality* than T if there exists an injection from T onto S, or $|S| \geq |T|$.
- S has *strictly higher cardinality* than T if it has a higher but not equal cardinality than T, or $|S| > |T|$ (i.e. we can inject T onto S but not the other way around).

- The idea of cardinality is essentially the generalization of the **size of a set**, but suited to categorize infinitely large and potentially uncountable set.
- Although two sets can be infinite, one's infinity/cardinality can be higher than the other's.

Cardinality**

Proposition 7

$\mathbb{R}$ is uncountable (i.e. $|\mathbb{R}| > |\mathbb{N}|$). However $|\mathbb{R}^n| = |\mathbb{R}|$ for any $n \in \mathbb{N}$.

I will prove the first part only. Clearly we can inject $\mathbb{N}$ onto $\mathbb{R}$ as $\mathbb{N} \subset \mathbb{R}$. To show the converse is not true, we only need to show that $(0, 1)$ cannot be injected onto $\mathbb{N}$.

Suppose that you could list out $(0, 1) = \{x_1, x_2, x_3, \dots\}$ Note any $\mathbb{R}$ can be represented as an (infinite) sequence, through its decimals. For example, represent 0.5 through $x = (5, 0, 0, 0, \dots)$ or $\pi - 3 = 0.14158\dots$ as $x = (1, 4, 1, 5, 8, \dots)$. In fact, $(0, 1)$ must contain every possible sequence with numbers $\{0, 1, 2, \dots, 9\}$ by construction.

Let's build an real number x^* so that its i-th element is any natural between 0 and 9 expect for the i-th element of x_i. Clearly, x^* is well-defined and belongs to $(0, 1)$. However, by the contradiction hypothesis, it must be that $x^* = x_i$ for some $i \in \mathbb{N}$, which implies $x_i^* = x_i^i$, which contradicts the definition of x^* .

Cardinality**

- The next result is crazy, but also a crucial property of the real number line. I will leave it up to you to prove it at home, but it is beyond the scope of the course

Exercise 9

At home, show that any interval in $\mathbb{R}$ has the same cardinality as $\mathbb{R}$

- So even the smallest interval you can think of $\mathbb{R}$ is, in a way, just as dense and large as the entirety of all numbers.
- In fact, if the interval is open, it is essentially the same as the real line as it also has no end.

3. Topology

Metric Spaces

Definition 18 (Metrics)

Let X be a set. A function $d : X^2 \to \mathbb{R}_+$ is a **distance function**, or **metric**, on X iff it satisfies the following properties:

1. $d(x, y) = 0$ iff $x = y$,
2. Symmetry: $d(x, y) = d(y, x)$, $\forall\, x, y \in X$, and
3. Triangle inequality: $d(x, y) \leq d(x, z) + d(z, y)$, $\forall\, x, y, z \in X$

If d is a metric on X, then the couple (X, d) is called a **metric space**.

- The distance function d is an inseparable part of a metric space, which is always a (set,function) couple.
- Elements in a metric space are often called **points**.
- Intuitively, a metric is a function that measures some notion of difference between two points.

Metric Spaces

1. The *standard* metric in $\mathbb{R}^k$ is the **Euclidean metric**:

$$d_2(x,y) := \sqrt{\sum_{i=1}^{k}(x_i - y_i)^2}$$

2. A generalization is the d_n metric in $\mathbb{R}^k$:

$$d_n(x,y) := \left(\sum_{i=1}^{k} |x_i - y_i|^n\right)^{\frac{1}{n}}, \quad n \in \mathbb{N}$$

3. The **discrete metric** in $\mathbb{R}^k$ is $d(x,y) = 1(x \neq y)$

Exercise 10

At home, try to show that these functions are all metrics in $\mathbb{R}^k$. (Hint: Look up the Cauchy-Schwarz and Minkowski inequalities.)

Open and Closed Sets

Definition 19 (Open ball)

Let (X, d) be a metric space. The **open ball** centered at $x \in X$ with radius $r > 0$ (or the r-**neighborhood** of x) is defined as the set

$$B_r(x) := \{z \in X : d(z, x) < r\}$$

- An open ball $B_r(x)$ depends both on both X and d
- In the metric space $(\mathbb{R}, d_2)$, the open ball $B_1(0) = (-1, 1)$
- In $(\mathbb{R}_+, d_2)$, the open ball $B_1(0) = [0, 1)$.
- For the discrete metric d, in $(\mathbb{R}, d)$ the open ball $B_1(0) = \{0\}$.

Definition 20 (Bounded Set)

Let (X, d) and $S \subset X$. We say that S is **bounded** iff there exists $x \in X$ and $r > 0$ such that $S \subset B_r(x)$.

Open and Closed Sets

Definition 21 (Interior point and Open Set)

Let (X, d) be a metric space, and S a subset of X. A point $x \in X$ is an **interior point** of S iff $\exists\, r > 0$ such that $B_r(x) \subset S$. The set of interior points of S is denoted as $\text{int}(S)$. The set S is an **open set** iff $S \subset \text{int}(S)$, i.e. all points in S are interior points.

- $S \subset \text{int}(S)$ is equivalent to $S = \text{int}(S)$.
- The empty set $\emptyset$ is open, because $\text{int}(\emptyset) = \emptyset$.
- Also, the whole space X is also open, because $\text{int}(X) = X$.
- $[0, 1)$ is not an open set in $(\mathbb{R}, d_2)$ as $0 \notin int([0, 1))$.
- $[0, 1)$ is an open set in $(\mathbb{R}_+, d_2)$: Note $B_{1/2}(0) = [0, 1/2) \subset [0, 1)$.

Proposition 8

In metric space (X, d), any open ball is an open set.

Open and Closed Sets

Proposition 9

In metric space (X, d):

1. Let $\{E_\alpha\}_{\alpha \in A}$ be an arbitrary family of open sets (potentially uncountably many of them). Then their union $\bigcup_{\alpha \in A} E_\alpha$ is also open.
2. Let $\{E_i\}_{i=1}^n$ be a finite family of open sets. Then their intersection $\bigcap_{i=1}^{n} E_i$ is also open.

Exercise 11

Show that the intersection of an countably infinite family of open sets may not be an open set.

Open and Closed Sets

Definition 22 (Limit point and Closed Sets)

Let (X, d) be a metric space, and $S \subset X$. A point $x \in X$ is a **limit point** of S iff $(B_r(x) \setminus \{x\}) \cap S \neq \emptyset$, $\forall\, r > 0$. The set of limit points of S is denoted as S'. We say S is a **closed set** iff $S' \subset S$.

- Intuitively, x is limit point if *any* open ball around x has points in S *other than x itself*.
- This time $S' \subset S$ does not imply $S' = S$.
- For example, in $(\mathbb{R}, d_2)$, for $S = [0, 1] \cup \{2\}$ we have $S' = [0, 1]$.
- $\emptyset$ and X are always closed sets of (X, d).
- $(0, 1]$ is not a closed set in $(\mathbb{R}, d_2)$ as 0 is limit point.
- $(0, 1]$ is a closed set in $(\mathbb{R}_{++}, d_2)$ as 0 is no longer a limit point.

Open and Closed Sets

Proposition 10

Consider metric space (X, d), $S \subset X$. The following are equivalent:

1. S is an open set.
2. S^c is a closed set.

Note that sets may be neither closed nor open!

Proposition 11

Consider metric space (X, d), and $S \subset X$

1. Let $\{F_\alpha\}_{\alpha \in A}$ be an arbitrary family of closed sets (potentially uncountably many of them). Then $\bigcap_{\alpha \in A} F_\alpha$ is also closed.
2. Let $\{F_i\}_{i=1}^n$ be a finite family of closed sets. Then $\bigcup_{i=1}^n F_i$ is also closed.

Open and Closed Sets

To make sure you understand open and closed sets *given the space you are dealing with*, try to solve the following.

Exercise 12

At home, prove the following properties of each set in each metric space (O = Open; NO = Not open; C = Closed; NC = Not Closed)

	$[0, +\infty)$	$(0, +\infty)$	$\{1/n : n \in \mathbb{N}\}$
In $(\mathbb{R}, d_2)$	NO and C	O and NC	NO and NC
In $(\mathbb{R}_+, d_2)$	O and C	O and NC	NO and NC
In $(\mathbb{R}_{++}, d_2)$	Not in space	O and C	NO and C

4. Sequences

Sequences

Definition 23 (Sequences)

Let X be a set. A **sequence** on X is simply a function $x : \mathbb{N} \to X$, usually denoted (x_n), or $(x_n)_{n\in\mathbb{N}}$

- Note that sequence and countably infinite sets are not the same thing.
- For example $(x_n) = (1_n) = (1, 1, 1, \dots)$ is a sequence on set $\{1\}$.
- Curly brackets $\{\}$ denote sets; parenthesis () denote sequences.
- A sequence is **(strictly) increasing** if $x_{n+1}(>) \geq x_n$ for any $n \in \mathbb{N}$. **(Strictly) Decreasing** sequences (strictly) decrease in n, and **monotone** sequences are either increasing or decreasing.
- A **subsequence** of (x_n) is a sequence that can be derived by deleting some of its elements. More formally, given a $(n_k)_{k\in\boldsymbol{N}}$ strictly increasing sequence in $\mathbb{N}$, the subsequence (x_{n_k}) is the composition

$$(x_{n_k})_{k\in\mathbb{N}} := (x_{n_k}, x_{n_2}, x_{n_3}, \dots) := ((x \circ n)_k)_{k\in\mathbb{N}}.$$

Sequences

Definition 24 (Limit)

Let (X, d) be a metric space. A sequence (x_n) in X is said to **converge to a point** $x \in X$, or **convergent**, iff $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $d(x_n, x) < \varepsilon$ for all $n > N$. Point x is called the **limit** of (x_n), and we denote it $x_n \to x$ or $\lim_{n\to\infty} x_n = x$.

Note that $d(x_n, x) < \varepsilon \equiv x_n \in B_\varepsilon(x)$. In words, convergence is the same as saying that the sequence eventually fits within any neighborhood of x, *no matter how small it is*.

Definition 25 (Bounded)

A sequence (x_n) on metric space (X, d) is said to be **bounded** if its range $\{x_1, x_2, \dots\}$ is a bounded set in (X, d).

Note that the range of (x_n) may not be the same as whole space X.

Sequences

Consider metric space (X, d), subset $S \subset X$

Proposition 12

The following three statements are equivalent:

1. S is a closed set.
2. (Sequential definition of closedness) For any sequence (s_n) in S that converges to some $s \in X$, we have $s \in S$.
3. S^c is an open set.

Proposition 13

If (x_n) in X is a converging sequence, then

1. The limit is unique.
2. (x_n) is bounded.

Sequences

Hence, let d_2 denote the Euclidean metric, $\leq$ the standard partial order.

Proposition 14

Let (x_n) and (y_n) be converging in $(\mathbb{R}, d_2)$ to x and y, respectively. If $x_n \leq y_n$ for any $n \in \mathbb{N}$, then $x \leq y$.

We say that the relation $\leq$ is preserved in the limit.

Exercise 13

At home, show that $\leq$ is preserved in the limit but $<$ is not.

Proposition 15

Let $(\boldsymbol{x}_n)$ be a sequence in $(\mathbb{R}^k, d_2)$. $(\boldsymbol{x}_n) \to \boldsymbol{x} = (x^1, x^2, \ldots, x^k)$ iff (x_n^i) converges to x^i in $(\mathbb{R}, d_2)$ for any $i \in \{1, 2, \ldots, k\}$.

Sequences

Proposition 16

Let (x_n) and (y_n) be converging in $(\mathbb{R}, d_2)$ to x and y , respectively. Then

(1) $x_n + y_n \to x + y$

(2) $x_n y_n \to xy$

(3) If $x \neq 0$, then $1/x_n = 1/x$

- That is, basic operations $+, \cdot$ and inversion are preserved under limit.
- Naturally substraction and division will also be preserved.

Exercise 14

1. Prove (1) and (3) in the above proposition.
2. Then prove that for the two converging sequences, $x_n - y_n \to x - y$ and $x_n / y_n \to x/y$ (if $y \neq 0$).

Sequences

Theorem 2 (Monotone Convergence)

Every increasing (decreasing) and bounded from above (below) real sequence (x_n) is convergent in $(\mathbb{R}, d_2)$.

Proposition 17

Every sequence in $(\mathbb{R}, d_2)$ has a monotone subsequence.

Theorem 3 (Bolzano-Weierstrass)

Every bounded sequence in $(\mathbb{R}^k, d_2)$ has a convergent subsequence.

The Bolzano-Weierstrass Thm is a very powerful and surprising result: From any sequence you can think of, even one fully random, as long as it is bounded, you can extract a converging subsequence!

5. Compactness

Compactness

Definition 26 (Open covers)

Let (X, d) be a metric space, $S \subset X$. A family of open sets $\mathcal{U} := \{E_\alpha\}_{\alpha \in A}$ in X is an **open cover** of S if $S \subset \bigcup_{\alpha \in A} E_\alpha$. Moreover a **finite subcover** of $\mathcal{U}$ is a finite subset of $\mathcal{U}$ that is also covering S.

A few examples to clarify this concept in $(\mathbb{R}, d_2)$

- $\mathcal{U}_1 = \{(0, 0.51), (0.49, 1)\}$ is an open cover of $(0, 1)$.
- $\mathcal{U}_2 = \{(-1, 1 - 1/n) : n \in \mathbb{N}\}$ is an (infinitely large) open cover $[0, 1)$.
- $\mathcal{U}_3 = \{(-0.01, 1.01 - 1/n) : n \in \mathbb{N}\}$ is an open cover of $[0, 1]$.
- $\mathcal{U}_4 = \{(-0.01, n) : n \in \mathbb{N}\}$ is an open cover of $[0, +\infty)$.

Exercise 15

Find a finite subcover of $\mathcal{U}_3$. Show $\mathcal{U}_2$ and $\mathcal{U}_4$ have no finite subcover.

Compactness

Definition 27 (Compactness)

Let (X, d) be a metric space, $S \subset X$. We say that S is **compact** if any open cover of S has some finite subcover.

The notion of compactness is tightly linked to closedness and boundedness. This is illustrated in examples from the previous slide:

- $[0, +\infty)$ is not compact because, although it is a closed set, it *had no end due to being unbounded*. That is why we were able to find a cover $\mathcal{U}_4$ that had no finite subcover.
- $[0, 1)$ is not compact because, although it is bounded, it *has holes in it due to not being closed* (the limit point 1 is not contained in it). Thus we were able to find cover $\mathcal{U}_2$ that admitted no finite subcover.
- Intuitively, $[0, 1]$ will be closed because it *has a clear boundary to it, it contains those boundaries and has no hole*. Proof is trickier though.

Compactness

Let's show some key results in Euclidean metric spaces

Proposition 18

Any closed interval $[a, b]$ is compact in $(\mathbb{R}, d_2)$.

This extends to $\mathbb{R}^k$:

Proposition 19

Every k-cell $[a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_k, b_k]$ is compact in $(\mathbb{R}^k, d_2)$.

Compactness

A few reason on why we prefer to deal with compact sets than closed ones.

Theorem 4 (Shrinking/Enlarging)

Let (X, d) be a metric space, $S \subset Y \subset X$. Then S is compact in (X, d) iff it is compact in (Y, d).

- We won't prove this, but it tells us that enlarging/shrinking the metric space will not change the compactness property of a set.
- Moreover, by making $Y = S$, the Proposition tells us that S is compact in (X, d) iff it is first of all compact in its own metric space (S, d). We say here that (S, d) is a **compact metric space**.
- Use this proposition to enlarge or shrink your metric space as needed and make it easier to prove a set is compact.

Compactness

Proposition 20

If S is compact in (X, d), then it is closed and bounded in (X, d) as well.

So compactness is a refinement of closedness and will inherit all of its properties, as well as being bounded.

Proposition 21

Let $S \subset Y \subset X$. If S is closed in (X, d) and Y is compact, then S is compact in (X, d).

This makes it easier to prove compactness, as you can show S is closed instead and contained in an obviously compact set. The two propositions imply that in compact metric spaces, closedness and compactness are equivalent.

Compactness

As you have noticed, compactness is a notoriously difficult definition to manipulate and use effectively. Thankfully, compactness = closedness + boundedness in Euclidean metric spaces:

Theorem 5 (Heine-Borel)

In $(\mathbb{R}^k, d_2)$ a set S is compact iff it is closed and bounded.

- In general metric spaces, a closed and bounded set may not be compact!
- In $(\mathbb{R}_{++}, d_2)$, we know $(0, 1]$ that it is bounded (obvious) and closed (since $0 \notin \mathbb{R}_{++}$).
- However it is not compact, since it obviously fails being compact in the $(\mathbb{R}, d)$ using Heine-Borel, and the Shrinking Theorem tells us that it will also fail compactness $(\mathbb{R}_{++}, d)$.

Compactness

There is also another friendlier characterization of compactness using sequences.

Definition 28 (Sequential Compactness)

Let (X, d) be a metric space, $S \subset X$. We say S is **sequentially compact** iff any sequence (x_n) in S has a subsequence converging to some $x^* \in S$.

This is a very useful property to know, but it turns out that this is equivalent to regular compactness in general metric spaces (but we will prove it only for Euclidean metric sapces).

Proposition 22

Let (X, d) be a metric space, $S \subset X$. It holds that S is compact in (X, d) iff it is sequentially compact in (X, d).

6. Continuity

Limits

In order to define continuity, we must first define a limit of a function

Definition 29 (Limit of a function)

Let (X, d_X) and (Y, d_Y) be metric spaces, $f : X \to Y$. We say that point $y_0 \in Y$ is a **limit of f at $x_0 \in X$** iff for any $\varepsilon > 0$, there exists some $\delta > 0$ such that

$$f(B_\delta(x_0) - \{x_0\}) \subset B_\varepsilon(y_0).$$

In this case we denote $\lim_{x \to x_0} f(x) = y_0$.

- Sometimes, the limit of f at x_0, and the actual value of f at x_0 may not coincide. Later on, we will call this a **discontinuity**.
- If it exists, the limit of f at x_0 *is unique*.

Limits

The *topological* definition of a function's limit from the last slide is often very difficult to handle. Thankfully, there is a sequence based characterization that we will use more frequently.

Theorem 6 (Sequence characterization of function limits)

Let (X, d_X) and (Y, d_Y) be metric spaces, $f : X \to Y$. Then $y_0 \in Y$ is a limit of f at $x_0 \in X$ iff for any sequence (x_n) in $X - \{x_0\}$ that converges to x_0, the sequence $(f(x_n))$ converges to y_0.

Since all sequence limits are unique, all function limit must also be unique.

Limits

Since limits of functions can be expressed as limits of sequences, they inherit many properties we learned prior.

Proposition 23

Let (X, d_X) be a m.s., $x_0 \in X$, $f : X \to \mathbb{R}^k$ with $f(\boldsymbol{x}) = (f_1(\boldsymbol{x}), f_2(\boldsymbol{x}), \ldots, f_k(\boldsymbol{x}))$. Then $\lim_{x\to x_0} f(x)$ exists iff $\lim_{x\to x_0} f_i(x)$ exists for any $i = 1, 2, \ldots, k$, and further, the i-th coordinate of $\lim_{x\to x_0} f(x)$ equals $\lim_{x\to x_0} f_i(x)$.

Proposition 24

Let (X, d_X) be a m.s., $f, g : X \to \mathbb{R}$ with limits at $x_0 \in X$. Then:

1. $\lim_{x\to x_0}[f(x) + g(x)] = \lim_{x\to x_0} f(x) + \lim_{x\to x_0} g(x)$.
2. $\lim_{x\to x_0}[f(x)\, g(x)] = \lim_{x\to x_0} f(x) \cdot \lim_{x\to x_0} g(x)$.
3. $\lim_{x\to x_0}(1/f) = 1/\lim_{x\to x_0} f(x)$, if $\lim_{x\to x_0} f(x) \neq 0$.

Continuity

Now let's define what continuity means, first using topology.

Definition 30 (Continuity of a function)

Let (X, d_X), (Y, d_Y) be m.s., $f : X \to Y$. We say that f is **continuous** at $x_0 \in X$ iff for any $\varepsilon > 0$, there exists some $\delta > 0$ such that

$$f(B_\delta(x_0)) \subset B_\varepsilon(f(x_0)).$$

We say f is a **continuous function** iff f is continuous in every point of its domain.

- Fun fact: If the domain some isolated points in their domain (e.g. $X = [0,1] \cup 2$) then is trivially continuous at those isolated points.

Continuity

- Intutitively, people often say a function is continuous when *small changes in inputs lead to similarly small changes in output*.
- A discontinuous function breaks this, because we expected a smooth/predictable behavior when reaching the discontinuity point, but the function took a discontinuous change there.
- Clearly the notion of a limit is related to continuity word-by-word. We state this formally below.

Theorem 7 (Sequence characterization of function continuity)

Let (X, d_X), (Y, d_Y) be m.s., $x_0 \in X$, $f : X \to Y$. We have f is continuous at x_0 iff for any sequence (x_n) in X that converges to x_0, we have $f(x_n) \to f(x_0)$.

In other words f is continuous iff $\lim_{x \to x_0} f(x) = f(x_0)$.

Continuity

Using the sequence characterization, the following properties can be quickly obtained as corollaries.

Proposition 25

Let (X, d_X) be a m.s., $x_0 \in X$, $f : X \to \mathbb{R}^k$. Then f is continuous at x_0 iff $f_i : X \to \mathbb{R}$ is continuous for any $i = 1, 2, \ldots, k$.

Proposition 26

Let (X, d_X) be a metric space, $x_0 \in X$, $f, g : X \to \mathbb{R}$ s.t. are continuous at x_0. Then

1. $f + g$ is continuous at x_0.
2. $f \cdot g$ is continuous at x_0.
3. $1/f$ is continuous at x_0 provided $f(x_0) \neq 0$.

Continuity

Another important result is that compositions of continuous functions are generally continuous

Theorem 8 (Continuity of compositions)

Let (X, d_X), (Y, d_Y), (Z, d_Z) be metric spaces, $f : X \to Y$ and $g : Y \to Z$ are continuous functions. Then $g \circ f : X \to Z$ is a continuous function.

- You all know that canonical functions $ln(x)$, x^α, e^x and $\sin x$ are continuous.
- Then using the last theorem and the last propostion from the previous slides, we get that basically any function constructed using sums, addition, multiplication, division and/or composition of the canonical function will be continuous.

Continuity

A second topological characterization of continuity exists which is as follows.

Theorem 9 (Second topological char. of continuity)

Let (X, d_X), (Y, d_Y) be metric spaces, $f : X \to Y$. We have f is continuous iff $f^{-1}(E)$ is open in (X, d_X) for any open set E in (Y, d_Y).

- In words, f is continuous iff the pre-image of an open set is also open.
- Although this definition is difficult to visualize compared to sequence or even the first topological definition, this is nonetheless a very useful result on its own that is good to keep in mind.
- In fact, we will use it to prove the famous Weierstrass Theorem.

Continuity

Theorem 10 (Weierstrass)

Let (X, d_X) be a metric space, $f : X \to \mathbb{R}$ is continuous using (X, d_X) and $(\mathbb{R}, d_2)$, and S is compact in (X, d_X). Then there must exist $x^*, x_* \in S$ s.t. $f(x^*) \geq f(x) \geq f(x_*)$ for any $x \in S$.

- This is extremely useful in economic theory, as many optimization problems we employ satisfy its pre-conditions and therefore, the existence of minima and maxima are guaranteed.
- To prove this we will proceed in two steps
  1. For any continuous function $f : X \to Y$, show that if some set K is compact in (X, d_X), then $f(K)$ is compact in (Y, d_Y).
  2. Then, for a compact set K in $(\mathbb{R}, d_2)$, there must be some $x^*, x_* \in K$ s.t. $x^* \geq x \geq x_*$.

7. Completeness**

Cauchy Sequences**

Let's define a *weaker* notion of convergence that *sounds like convergence* but strictly speaking it is not.

Definition 31 (Cauchy Sequences)

In metric space (X, d), a sequence (x_n) is said to be (a) **Cauchy (sequence)** iff for any $\varepsilon > 0$, there exists some $N \in \mathbb{N}$ such that $d(x_m, x_n) < \varepsilon$ for any $m, n > N$.

- In words, a sequence is Cauchy if its points become progressively and infinitesimally closer together.
- The crucial difference with the regular convergence definition is that Cauchy sequences don't make any reference to a limit point $x \in X$.

Exercise 16

In $(\mathbb{Q}, d_2)$, show that $((1 + 1/n)^n)$ *does not converge* but is Cauchy.

Cauchy Sequences**

Cauchy and convergent sequences do share a lot of properties.

Exercise 17

(At home) show that a Cauchy sequence in any metric space (X, d) must be bounded.

Most importantly, a convergent sequence is formally a stronger notion of Cauchy sequence

Proposition 27

If a metric space (X, d), if (x_n) is convergent then it is Cauchy.

The reason why we care about this notion is that it is an *auxiliary concept* that is useful to define the notion of a *complete space*.

Completeness**

Let's define an additional topological notion related to closedness and compactness; in fact it is a *middle point* between the two.

Definition 32 (Completeness)

Let (X, d) be a metric space, $S \subset X$. We say that S is **complete** iff any Cauchy sequence in S converges to a limit point in S. Further, if X is complete in (X, d), we say (X, d) is a **complete metric space**.

- Intuitively, a complete set is one that *has no holes in it*.
- For example, following on the last slide, $(\mathbb{Q}, d_2)$ is not complete.
- In fact, the set of irrationals $\mathbb{R} - \mathbb{Q}$ is not complete either. Just take sequence $(\pi/n)_{n\in\mathbb{N}}$.

Completeness**

Proposition 28

Let (X, d) be a metric space, $S \subset X$. If S is compact, then it is complete. And if S is complete, then it is closed.

- Importantly, complete sets do not need to be compact, since they may be unbounded.
- $(\mathbb{R}, d_2)$ is the canonical complete set, clearly it is not compact.

Proposition 29

Euclidean spaces $(\mathbb{R}^k, d_2)$ are complete metric spaces.

Fixed points**

- We are learning of completeness because they are the minimum topological requirement to achieve the famous Contraction Mapping Theorem, which appears often in problems of recursive nature.
- To learn this theorem, first let's define what a fixed point is.

Definition 33 (Fixed point)

Consider a **self-map** f, that is, a function that maps from the domain onto itself, or $f : X \to X$. A point $x^* \in X$ is said to be a fixed point iff $f(x^*) = x^*$.

- E.g., $f : [0, 1] \to [0, 1]$ with $f(x) = x^{0.5}$ has fixed points 0,1.
- In a self map it makes sense to talk about f^n which is just the composition of f with itself n times.

Fixed points**

- More generally, you will find in applications that a fixed point represents a notion of *convergence* or *equilibrium* for some operation/algorithm/model.
- For example, suppose that two people, a buyer b and seller s, are negotiating the price for the sale of a good and function $x_b = f(x_s)$ tells you the ask price of the buyer x_b when seller's ask price was x_s.
- A fixed point for f would be an equilibrium (x^*, x^*) at which the ask price for the buyer is the same at which the seller wants to sell it for. The two people have reached an agreement for the price.

Contractions**

Now let's learn a specific type of self-map

Definition 34 (Contractions)

Let (X, d) be a metric space. A self-map $f : X \to X$ is said to be a **contraction** if there exists some non-negative real number $0 \leq \lambda < 1$ such that

$$d(f(x), f(x')) \leq \lambda \cdot d(x, x')$$

for any $x, x' \in X$.

- For example, $f(x) = x^{0.5}$ is not a contraction. Just pick $x = 0, x' = 1$
- However $f(x) = 0.5x$ is a (trivial) contraction.

Contractions**

Theorem 11 (Contraction Mapping)

Let (X, d) be a complete metric space, and $f : X \to X$ be a contraction. Then f has a unique fixed point x^*. Further, for any $x \in X$, we have that $\lim_{n\to\infty} f^n(x) = x^*$.

- For example, with $f(x) = 0.5x$, clearly 0 is a fixed point.
- Moreover $f^n(x) = 0.5^n x$ which converges to 0 for any x.
- To prove this, follow these steps:
  1. Show that $(f^n(x_0))$ is Cauchy for any $x_0 \in X$. By completeness of (X, d), this implies $(f^n(x_0))$ converges to some $x^* \in X$.
  2. Show that $x^* := \lim_{n\to\infty} f^n(x_0)$ is a fixed point of f.
  3. Show that the fixed point must be unique.